



Determinants of Individual Investment in Green Financial Instruments: Evidence from a Bias-Corrected Logistic Model

Alan L. Sampaolo

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Abstract

This project examines the factors influencing individual investment in green financial instruments, based on survey data from 97 Master's students at the University of Pavia. To address the small sample size and the occurrence of rare events, the analysis applies logistic regression models with Firth bias correction. Two dependent variables are considered: **y.ipo**, capturing the stated intention to invest in green instruments, and **y.real**, reflecting the revealed choice of respondents who have already allocated part of their capital to such products. This dual perspective enables a comparison between what individuals say they would do and what they actually decide when faced with a concrete investment scenario.

The results highlight a notable divergence. Stated intentions appear mainly associated with financial readiness: respondents willing to allocate a substantial share of a hypothetical unexpected windfall to financial instruments in general are also more likely to express interest in green products. Conversely, revealed preferences, although statistically limited by sample size, suggest that environmental concern may play a stronger role in actual investment behavior.

Overall, the findings illustrate the dual nature of sustainable investment decisions. Survey declarations seem to mirror economic confidence and investment culture, whereas real behavior may be more strongly influenced by intrinsic environmental values. Given the exploratory scope and limited dataset, the study should be regarded as an initial step toward understanding the behavioral foundations of sustainable finance and as a basis for future research on more diverse and larger samples.

Contents

1	Introduction	3
2	Data and Preprocessing	4
3	Methodology	6
3.1	Estimation Strategy	6
3.2	Model Evaluation	6
3.3	Interpretation of Coefficients	6
4	Results	7
4.1	Firth Logistic Performance – Non-aggregated Dataset	7
4.1.1	Stated Preferences (y.ipo)	7
4.1.2	Revealed Preferences (y.real)	8
4.2	Sample Size Sensitivity	8
4.3	Firth Logistic Performance – Aggregated Dataset	9
4.3.1	Stated Preferences (y.ipo)	9
4.3.2	Revealed Preferences (y.real)	9
5	Discussion and Limitations	11

1 Introduction

Over the past decade, sustainable finance has gained increasing prominence among policymakers, financial institutions, and investors. The growing market for Environmental, Social and Governance (ESG) products, including green bonds, ESG exchange-traded funds, and sustainable mutual funds, reflects an ongoing effort to align capital allocation with the transition toward a low-carbon economy. In Europe, initiatives such as the EU Taxonomy and the Sustainable Finance Disclosure Regulation have further strengthened the role of financial markets in supporting environmentally responsible projects.

Despite this institutional momentum, the participation of individual investors remains uneven. While the determinants of institutional demand and asset pricing have been extensively studied, considerably less attention has been devoted to understanding the factors shaping individual-level engagement in sustainable finance.

Moreover, previous research based on survey data suggests a possible divergence between stated intentions (what respondents claim they would do) and revealed behavior (what they actually do when facing a real allocation decision). Such discrepancies may stem from social desirability bias, whereby individuals express socially valued attitudes without necessarily reflecting them in actual behavior. This makes it essential to compare stated and revealed preferences within a controlled empirical framework.

The questionnaire used in this study collected a wide range of categorical variables describing financial behavior, environmental attitudes, and perceptions of green products. Rather than analyzing each variable in isolation, these indicators can be conceptually grouped into **four overarching dimensions**, each representing a potential barrier that might discourage individuals from investing in green financial instruments:

- **Limited capital availability:** low ability to allocate funds to sustainable products.
- **Low investment propensity:** reluctance to invest in financial instruments, even when additional disposable income is available.
- **Low environmental sensitivity or information:** weak awareness of ecological issues, limited willingness to trade off returns for sustainability, or lack of knowledge about international environmental agreements.
- **Distrust in green products:** perception that these instruments have limited real impact or are no safer than traditional ones.

Interpreting the dataset through these four dimensions provides a structured framework to identify the main obstacles preventing young investors from engaging in sustainable finance, while also offering a foundation for designing interventions to address them.

From a methodological perspective, the study applies logistic regression models to binary outcomes. Since the dataset is relatively small and includes rare events, Firth’s bias-reduced logistic regression is employed to mitigate estimation bias and handle potential separation issues [1].

The contribution of this project is threefold. First, it offers a methodological application of bias-corrected logistic regression in a small-sample, rare-event context. Second, it provides exploratory empirical insights into the drivers of sustainable investment at the individual level. Third, it highlights the divergence between stated and revealed preferences, contributing to the broader understanding of behavioral dynamics in sustainable finance.

The remainder of the paper is organized as follows. Section 2 describes the dataset and preprocessing steps. Section 3 outlines the methodology. Section 4 presents the empirical results, and Section 5 discusses the findings and limitations.

2 Data and Preprocessing

The empirical analysis is based on survey data collected from 97 Master’s students in Quantitative Finance at the University of Pavia. The questionnaire was designed to capture both individual characteristics and perceptions of product features relevant to sustainable investing. Two binary dependent variables were defined: **y.ipo**, representing the stated intention to invest in green financial instruments, and **y.real**, representing the revealed choice of respondents who already allocate part of their capital to such products. All remaining items were treated as independent variables.

All independent variables were recoded into binary dummies. **Table 3** in Appendix A reports the full coding scheme linking each dummy to the original survey question and response category.

Preprocessing. Several steps were undertaken to prepare the dataset for econometric analysis. A technical index column was first removed, and duplicate variables were checked (none detected). Missing values in both regressors and dependent variables were replaced with zeros, ensuring that the full sample of 97 observations was retained. No responses were excluded.

All categorical regressors were transformed into binary dummy variables, while both dependent variables were mapped into a 0/1 format.

Multicollinearity and redundancy. To prevent perfect collinearity, one baseline category was systematically dropped from each set of dummies.

The Pearson correlation matrix identified three variables with correlations above 0.8: *perc.inv.26.50*, *info.very*, and *paris.some*. The latter was dropped because it was perfectly correlated with *info.very*. Additionally, two adjacent categories were merged into a new dummy, *perc.inv.lt.26.50*, combining *perc.inv.lt.25* and *perc.inv.26.50*. Variables showing no variation (always equal to 0 or 1), such as *perc.inv.gt.75* and *perc.reg.75*, were also excluded since they provide no information and cause singularities in estimation.

Further checks using the `findLinearCombos` function in R confirmed and removed any remaining exact linear dependencies among regressors. These adjustments ensured that the final regressor set was free from redundancy and multicollinearity.

After preprocessing, the final dataset retained all 97 respondents and included 14 explanatory variables. Conceptually, these regressors can be grouped into the four behavioral dimensions introduced earlier: limited capital availability, low investment propensity, low environmental

sensitivity or information, and distrust in green products. This cleaned and structured dataset served as the foundation for the subsequent econometric analysis.

3 Methodology

The empirical strategy relies on binary logistic regression models estimated separately for the two dependent variables: **y.ipo** (stated intention) and **y.real** (revealed choice). Logistic regression is suitable for dichotomous outcomes and allows for a probabilistic interpretation of individual decision-making behavior.

3.1 Estimation Strategy

The estimation procedure was carried out in two stages.

Stage 1 – Non-aggregated dataset. Models were first estimated on the dataset obtained through the preprocessing described in **Section 2**, hereafter referred to as the *non-aggregated dataset*. Initial attempts using classical maximum likelihood estimation produced unstable results: coefficients diverged, standard errors were inflated, and complete separation issues emerged. Applying Firth’s bias-reduced logistic regression [1] successfully addressed separation but did not fully stabilize inference, as confidence intervals remained extremely wide and most p -values were close to one.

Stage 2 – Aggregated dataset. To mitigate these problems, a more aggregated dataset was constructed by merging rare dummy variables, defined as categories with fewer than 10% of observations in either state. This *aggregated dataset* reduced sparsity and improved balance among regressors. Firth logistic regression was then re-estimated, yielding more stable coefficient estimates and narrower confidence intervals.

3.2 Model Evaluation

Model performance was evaluated from both in-sample and out-of-sample perspectives.

In-sample evaluation. Discriminatory power was measured by the Area Under the Curve (AUC), calibration by the Brier score, and explanatory ability by Tjur’s R^2 .

Out-of-sample evaluation. Predictive robustness was assessed through stratified five-fold cross-validation.

3.3 Interpretation of Coefficients

Two complementary approaches were adopted for interpreting the results.

First, odds ratios (ORs) and their confidence intervals were reported to indicate whether each factor increases or decreases the likelihood of investing in green instruments.

Second, average marginal effects (AMEs), expressed in percentage points, were computed to provide a more intuitive measure of the change in predicted probability associated with a one-unit variation in each regressor.

Together, these measures allow both statistical and practical interpretation of the determinants of individual investment in green financial instruments.

4 Results

4.1 Firth Logistic Performance – Non-aggregated Dataset

The performance metrics for the Firth logistic models estimated on the *non-aggregated dataset* are presented first.

In-sample evaluation. Both models (**y.ipo** and **y.real**) yield an AUC of 1, very low Brier scores (≈ 0.005), and Tjur’s R^2 around 0.86.

Cross-validation. Five-fold stratified cross-validation produces nearly identical results, with mean AUC = 1 and minimal variability across folds, while the Brier scores and Tjur’s R^2 remain equally stable.

Although these metrics appear outstanding, they likely reflect quasi-perfect separation and overfitting rather than genuine predictive accuracy. Given the limited number of positive outcomes, particularly for **y.real**, the models can effectively memorize sample patterns, inflating performance measures. This highlights the need to interpret the coefficients with caution and to place greater emphasis on the aggregated specifications.

4.1.1 Stated Preferences (**y.ipo**)

Results for **y.ipo** indicate one clear driver of stated intentions: the share of a hypothetical unexpected windfall that respondents would allocate to financial instruments in general (not necessarily green).

Relative to the 0% baseline, respondents allocating 26–50% of such additional income are substantially more likely to state that they would invest in green instruments (OR ≈ 165 , $p < 0.001$; AME $\approx +45$ pp). The effect is even stronger for those indicating 51–75% (OR extremely large, $p \approx 0.045$; AME $\approx +51$ pp). The 1–25% group is marginally significant ($p \approx 0.057$) but still exhibits a considerable marginal effect (+53 pp).

Because the question referred to a hypothetical windfall (e.g., a graduation gift) rather than regular income, reluctance to invest cannot be attributed to strict financial constraints. A more plausible explanation is limited financial literacy: respondents unwilling to allocate even unexpected additional income likely reflect weaker familiarity with financial instruments and investment practices. This interpretation aligns with the observation that about 38% of respondents answered “no” to **y.ipo**, indicating no intention to invest in green products, an outcome that may partially stem from insufficient financial knowledge.

Other regressors display large point estimates but limited statistical reliability. For instance, *safe.little* (OR ≈ 237 , $p \approx 0.073$) and *imp.some* (OR ≈ 187 , $p \approx 0.084$) exhibit notable odds ratios and non-negligible marginal effects ($\approx +50$ pp).

Despite these large odds ratios, the wide confidence intervals and borderline significance levels suggest that results should be interpreted with caution. Variables linked to environmental sensitivity or safety perceptions generally display negative marginal effects, though none reach conventional significance thresholds.

Variable	OR	95% CI (low)	95% CI (high)	p-value
perc.reg.26.50	165.000	6.501	54485.444	0.0006
perc.reg.51.75	1.54e10	1.942	1.87e20	0.045
perc.reg.lt.25	1.08e5	0.630	2.58e10	0.057
safe.little	236.867	0.524	250101.827	0.073
imp.some	187.000	0.411	198616.615	0.084
env.some	0.000	0.000	18993.319	0.162
safe.some	0.001	0.000	171.980	0.219
env.little	0.000	0.000	24098.951	0.225
env.very	1086.579	0.003	4.17e8	0.246
info.some	173.702	0.004	8.72e5	0.262
perc.inv.51.75	919.413	0.002	1.03e9	0.276
info.little	658.246	0.000	5.16e9	0.398
paris.little	1.255	0.002	924.516	0.938
imp.little	1.133	0.006	222.371	0.952

Table 1: Firth logistic regression results for **y.ipo** (*non-aggregated dataset*).

Summary. In the *non-aggregated dataset*, stated intentions to invest in green instruments are primarily explained by willingness to allocate a substantial share of extra income to financial instruments in general, while other factors show limited and inconclusive evidence.

4.1.2 Revealed Preferences (y.real)

When moving from stated intentions (**y.ipo**) to revealed choices (**y.real**), the analysis becomes considerably less robust. None of the regressors achieves statistical significance, with confidence intervals spanning several orders of magnitude.

Average marginal effects confirm this instability. Although *env.very* suggests a potentially strong positive association ($AME \approx +78$ pp), the result remains highly uncertain and statistically inconclusive. Other regressors exhibit only minor variations in predicted probabilities, none of which can be interpreted with confidence.

Summary. The revealed-preference model based on the *non-aggregated dataset* does not yield robust evidence regarding the determinants of actual investment behavior. The apparent signal from *env.very* should be regarded as suggestive but not definitive.

4.2 Sample Size Sensitivity

Given the small sample size ($N = 97$) and the instability of inference, a simple bootstrap exercise was conducted on the *non-aggregated dataset* to evaluate how many observations would be required for selected regressors to achieve conventional significance (for the y.real model). The results indicate that even by doubling or tripling the sample (to about $N \approx 200$ –300), the coefficients would still fail to reach the 5% significance threshold consistently.

4.3 Firth Logistic Performance – Aggregated Dataset

The models estimated on the *aggregated dataset*, where rare dummies (fewer than 10% of ones or zeros) were merged, show a broadly similar pattern of performance.

In-sample evaluation. Metrics again indicate near-perfect discrimination, which should be interpreted as evidence of quasi-complete separation rather than genuine predictive accuracy.

Cross-validation. The cross-validation results confirm similar metrics across folds.

4.3.1 Stated Preferences (y.ipo)

The main finding from the *non-aggregated dataset* persists: respondents allocating 26–50% of an unexpected windfall to financial instruments remain significantly more likely to express an intention to invest in green products (OR ≈ 165 , $p < 0.001$; AME $\approx +48$ pp). By contrast, the 51–75% and 1–25% categories lose significance after aggregation, though their marginal effects remain positive. Beyond financial willingness, no other variables reach conventional significance levels. Regressors related to environmental attitudes or safety perceptions show large marginal effects but extremely wide confidence intervals, limiting their interpretability.

Summary. Compared to the non-aggregated case, aggregation stabilizes coefficient estimates but reduces the number of significant predictors. Stated preferences for green investment continue to be mainly explained by general investment readiness. Respondents indicating 0% allocation remain the least likely to express interest in green instruments.

4.3.2 Revealed Preferences (y.real)

The *aggregated dataset* provides slightly clearer, though still inconclusive, insights regarding revealed preferences. None of the regressors achieves significance at the 5% level.

The variable *env.very* (very high environmental concern) emerges as the closest to relevance (OR ≈ 132 , $p \approx 0.112$; AME $\approx +82$ pp), suggesting that strong environmental awareness may be positively associated with actual investment in green instruments. However, the evidence remains tentative given the wide confidence interval.

Other regressors, including willingness to allocate extra income and perceptions of safety or impact, remain statistically unreliable. Their odds ratios cluster around unity, confidence intervals remain wide, and marginal effects small.

Summary. In the *aggregated dataset*, revealed preferences show suggestive evidence that pro-environmental attitudes could play a role in actual investment behavior, although the results remain exploratory due to the limited sample size.

Variable	OR	95% CI (low)	95% CI (high)	p-value	AME (pp)
env.very	131.532	0.223	105640.370	0.112	82.342
safe.little	0.531	0.001	542.694	0.730	-3.421
paris.little	1.859	0.005	1092.536	0.772	3.309
imp.little	0.748	0.003	27.401	0.821	-1.593
perc.reg.lt.25	1.588	0.005	507.703	0.851	2.584
imp.some	0.703	0.000	982.357	0.876	-1.813
perc.reg.26.50	1.364	0.007	271.973	0.881	1.704
env.little	0.579	0.000	6399.798	0.892	-3.409
info.little	0.727	0.000	2324.254	0.918	-1.909
env.some	0.749	0.000	3291.444	0.943	-1.530
perc.inv.51.75	1.153	0.001	80985.775	0.962	0.804
safe.some	0.920	0.007	236.986	0.964	-0.448

Table 2: Firth logistic regression results for **y.real** (*aggregated dataset*).

5 Discussion and Limitations

The analysis highlights a clear contrast between stated and revealed preferences.

For stated intentions (**y.ipo**), the decisive factor is respondents' willingness to allocate a portion of an unexpected windfall to financial instruments in general. Students declaring a readiness to invest 1–25%, 26–50%, or 51–75% of such additional income show a much higher probability of responding “yes” to green investment compared with those allocating 0%, who serve as the baseline category. This pattern suggests that individuals unwilling to invest even a small share of extra income are also the least likely to express interest in green products. Because the question referred to a hypothetical gain rather than regular earnings, this reluctance cannot plausibly be attributed to financial necessity. Instead, it appears linked to limited financial literacy, a weak investment culture, and low confidence in financial products. Consistent with this interpretation, approximately 38% of respondents reported no intention to invest in green instruments, an outcome that can reasonably be associated, at least in part, with gaps in financial knowledge.

In contrast, the revealed-preference model (**y.real**) presents a different picture. None of the regressors reaches conventional significance, yet the variable capturing very high environmental concern (*env.very*) emerges as the closest to relevance. Its large marginal effect suggests that intrinsic environmental values may influence actual investment behavior, even though the statistical evidence remains too limited to draw firm conclusions.

Taken together, these findings imply that declarations and behavior capture two distinct dimensions of decision-making. Stated preferences appear to reflect economic readiness and financial literacy, whereas revealed preferences, although statistically fragile, point toward a greater role for environmental attitudes. This divergence underscores the importance of distinguishing between what individuals say they would do and what they actually choose when facing real financial decisions.

In summary, while the evidence remains exploratory, the results convey an important message: promoting sustainable investment requires progress on two complementary fronts, enhancing financial education to strengthen investment readiness, and nurturing environmental awareness to translate intentions into action.

Several limitations must nonetheless be acknowledged. First, the regressors exhibit some redundancy and multicollinearity, resulting in inflated standard errors and unstable inference, issues that cannot be fully resolved by simply increasing the sample size. Second, the dataset is relatively small ($N = 97$), with few positive outcomes for revealed preferences, which further limits statistical power and widens confidence intervals.

Overall, this study should be regarded as a pilot analysis. A more diversified and larger dataset, combined with a refined selection of explanatory variables, would provide a stronger foundation for policy and market insights on sustainable investment behavior.

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References

- [1] David Firth. Bias reduction of maximum likelihood estimates. *Biometrika*, 80(1):27–38, 1993.

Appendix A – Variable Coding

Survey question	Response category	Variable code
Q1: Percentage of own capital invested in financial instruments	0% 1–25% 26–50% 51–75% >75%	baseline perc.inv.lt.25 perc.inv.26.50 perc.inv.51.75 perc.inv.gt.75
Q2: Percentage of extra income (e.g. gift) willing to invest	0% 1–25% 26–50% 51–75% >75%	baseline perc.reg.lt.25 perc.reg.26.50 perc.reg.51.75 perc.reg.75
Q3: Environmental sensitivity	None Low Medium High	baseline env.little env.some env.very
Q4: Information on green initiatives	None Low Medium High	baseline info.little info.some info.very
Q5: Knowledge of the Paris Agreement	None Low Medium High	baseline paris.little paris.some paris.very
Q6: Belief that green investments have real impact	None Low Medium High	baseline imp.little imp.some imp.very
Q7: Perception of safety of green instruments with respect to traditional ones	None Low Medium High	baseline safe.little safe.some safe.very
Q8: Maximum annual return willing to sacrifice to acquire a green rather than a traditional financial instrument	0% 0–1% 1–2% 2–2.5% >2.5%	baseline max.lt.1 max.1_2 max.2_2.5 max.gt.2.5
Q9: Would you invest part of your capital in green instruments?	Yes/No	1/0
Q10: Do you already invest in green instruments?	Yes/No	1/0

Table 3: Coding of survey responses into dataset dummy variables